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Design and stability analysis of a control system for a grid-independent direct current microgrid with hybrid energy storage system

Prashant Singh *, J.S. Lather

Electrical Engineering Department, National Institute of Technology, Kurukshetra, Haryana 136119, India

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ABSTRACT

This paper proposes the design and stability analysis of a control system for a hybrid energy storage system (HESS). The control method is framed to address the demand-generation disparity and DC bus voltage regulation. In the proposed control method, battery is utilized to address slow-frequency power surges. Meanwhile, supercapacitor (SC) has been used to manage fast-frequency power surges. Furthermore, the proposed control method enhances the battery life cycle by diverting uncompensated battery currents to SCs for quick compensations. The advantages of the proposed control method over the conventional control method show its robustness for DC bus voltage in terms of settling time and voltage overshoot/undershoot. Additionally, total harmonic distortion (THD) of 3-phase voltage is presented. The test scenarios have been verified in a MATLAB/SIMULINK environment. The experimental results authenticate the viability and feasibility of the proposed control method.

1. Introduction

The need for distributed energy generation via notable improvements from renewable energy resources (RERs) has sparked increasing trends globally over the last decade. The development of distributed generation, containing one from RERs, is exciting because a significant field remains for energy distribution enrichment beyond current, long-distance, thermally accentuated transmission lines. Due to regional political matters, new transmission corridors are limited to the economic scenario and may need more time to complete. Apart from this, a microgrid is the best solution if the power generating station is far from a distant area and is more cost-effective due to its distributed generation functionality [1]. DC microgrids (DCMGs) are becoming increasingly popular and compatible as RERs, DC loads, and energy storage devices infiltrate enormously due to their remarkable flexibility, reliability, and performance [2,3]. Because of the irregular and stochastic nature of the RERs, the DCMGs relies on a battery storage system to achieve generation and demand [4,5]. Owing to the batteries' low energy densities, power demands cannot be fulfilled over an extensive range of operating conditions. Supercapacitor (SC) has a high power density that fills short-term power demand. However, it does not have prolonged resilience due to its low energy density [6]. Therefore, in this work, we consider the battery and SC's hybridization, known as hybrid energy storage system (HESS), to benefit from high power capacity and high energy density. The study in [7] broadly discusses the benefits of HESSs.

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* Corresponding author.
E-mail address: prashant_6170069@nitkkr.ac.in (P. Singh).

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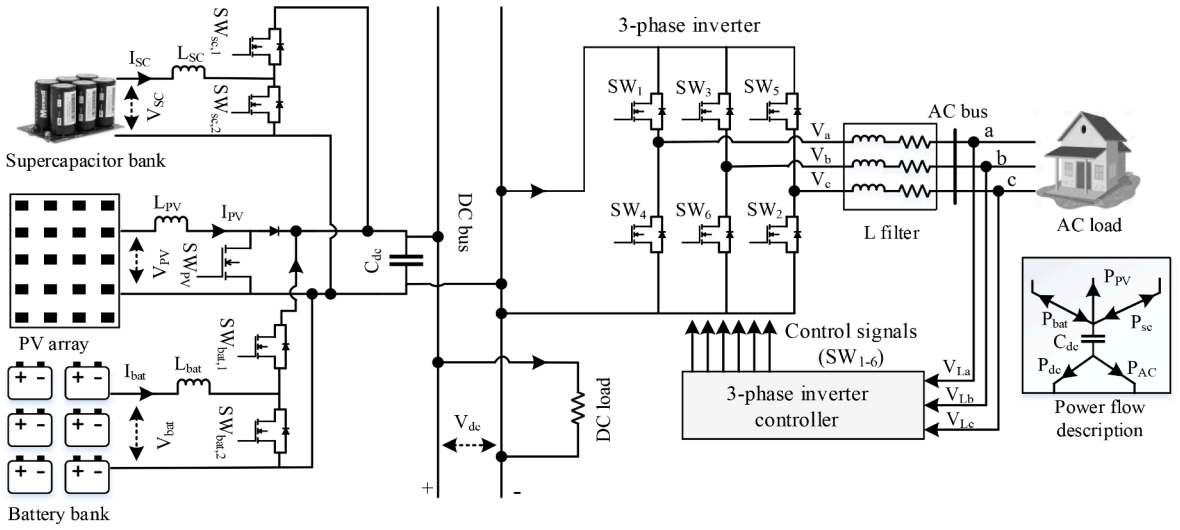


Fig. 1. System configuration of considered DCMGs where I_{sc} = Supercapacitor current (A), I_{pv} = PVA current (A), I_{bat} = Battery current (A), L_{PV} = PVA filter inductance (H), L_{bat} = Battery filter inductance (H), L_{sc} = Supercapacitor filter inductance (H), C_{dc} = DC bus filter capacitance (F), V_{dc} = DC bus voltage (V), SW_{PV} = PVA boost converter switch, $SW_{sc,1}$, $SW_{sc,2}$ = Supercapacitor bidirectional DC/DC converter switches, $SW_{bat,1}$, $SW_{bat,2}$ = Battery bidirectional DC/DC converter switches, SW_{1_6} = three-phase converter switches, P_{PV} = PVA output power (W), P_{bat} = Battery output power (W), P_{sc} = SC output power (W), P_{dc} = DC load power (W), P_{AC} = AC load power (W), v_a,$v_b$andv_c = three-phase inverter output voltages (V), v_{La},v_{Lb}andv_{Lc} = three-phase voltages (V), i_a,$i_b$andi_c = three-phase currents (A).

The research in [8] proposes a deadbeat-based power management approach for DCMGs, including HESS. The proposed approach regulates the DC bus voltages with a fast dynamic response during source and load changes. However, the generated reference signal becomes very high, so real hardware implementation is challenging, damaging the system and creating instability. The finding in [9] studies a model-predictive control-based power management approach for DC bus voltage regulation and maintaining battery and SCs State-of-charge (SoC) within the specified limit. The presented approach that is relies on model-predictive control, which requires complex mathematical computations. However, the study does not discuss issues related to microgrid stability with HESS. The study in [10] exhibits a distributed iterative learning mode for DCMGs that attempts to resolve the voltage control problem, despite not being locked into the proper parameters. The Bellman optimality theory guarantees that the proposed DCMG settles into a Nash equilibrium and maintains optimum voltage control. Moreover, [11] presents an iterative learning model for hybrid microgrids to regulate DCMG's current/power-sharing and the DC bus voltage under differing power generation and load conditions. However, differing power generation and load changes will increase the battery's stress, thus creating instability in the system. In [12] a distributed discrete-time algorithm is composed to control the bus voltage in DCMG. Additionally, a periodic event-triggered discrete-time approach is employed to reduce the controller's communications needs. However, communication channel requirements lead to extra costs, and the failure of any communication channel will stop control action. A distributed cooperative consensus approach [13] for HESSs is suggested that controls DC bus voltage and manages power-sharing in a DCMGs. This distributed recovery approach guarantees the system's effectiveness if the existing network is not connected or fails. A consensus-based algorithm [14] is elaborated to control the DCMGs bus voltage and maintain power splitting even among DCMGs components. A lemma is formulated to help instability analysis by exploiting the Laplacian matrices' properties. The Lyapunov study proved that voltage control errors converged to zero asymptotically. The study in [15] introduces virtual inertia regulation-based unified control framework for voltage control of DCMG to regulate the DC bus voltage and replicate inertia. Besides the efficiency improvement, a restoration control is usually employed to lessen the steady-state error in the DC bus voltage. In [16] introduces an artificial neural network-based power management approach for DCMGs, including HESSs, to ensure power-sharing and bus voltage regulation. In [17] presents a combined proportional integral sliding mode control method for DCMGs to control DC bus voltage during differing power generation and load changes. In [18] exhibits an energy management approach for HESS to monitor DC bus voltage and improve battery life by reducing stress on the battery. However, the power converter's lowered response time will place enormous stress on the battery, which will decrease the battery's charge/discharge rate. While the HESS is practiced in the literature, the modeling, configuration, and stability issues of DCMGs with these storage systems are not addressed adequately.

Considering these issues, this paper proposes the design and stability analysis of a control system for HESSs to maintain demand-generation disparity and DC bus voltage regulation, acknowledging changes in power generation and load. The contributions of this work are:

(1) Detailed modeling of DCMGs with HESSs.
(2) Improved DC bus voltage regulation under changes in generation and load.
(3) Effective power-sharing within HESSs for considered events.

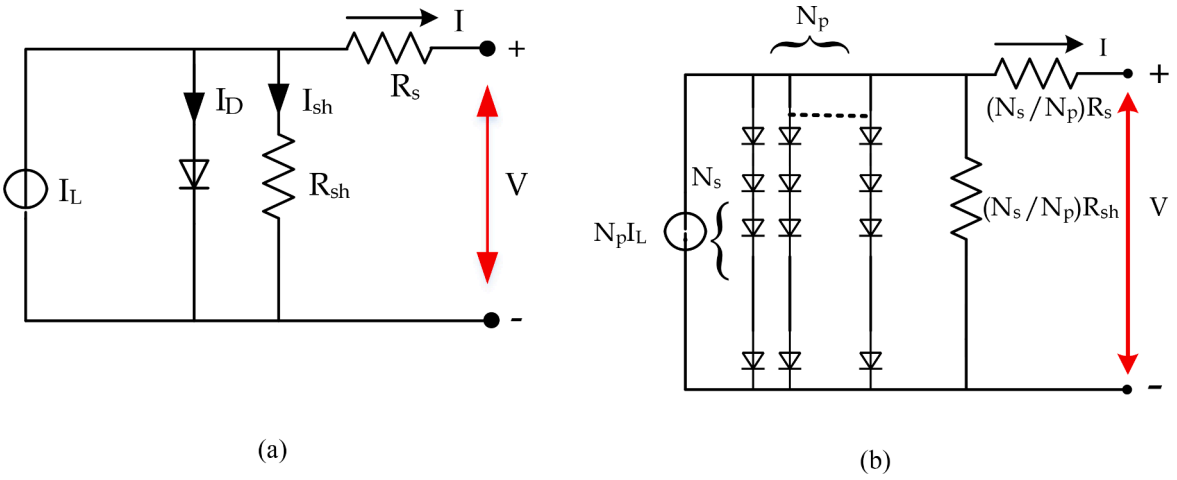


Fig. 2. (a) Circuit diagram of single diode PV cell (b) PV module.

The article is structured as follows: Section 2 illustrates the DCMG system configuration, including component modeling. Section 3 details the design and stability analysis of the proposed control method. Section 4 exhibits the simulations, including the proposed control method of the studied DCMG system. Additionally, the section validates the proposed method experimentally on a real-time platform using hardware-in-loop (HIL) simulator. Section 5 highlights the conclusions.

2. DCMGs configuration and modeling

Fig. 1 illustrates a DCMG comprised of photovoltaic array (PVA), HESSs, DC load, and AC load. The PVA is combined with DC bus via boost converter, while the battery and SC are combined to the DC bus via bidirectional DC/DC converters (BDDCs). Simultaneously, the AC load is associated with AC bus. As load demand increases, the battery regulates the DC bus voltage while discharging. Moreover, in PV power generation exceeding load requirement, the battery regulates the DC bus voltage and charge the HESSs.

2.1. Modeling of solar PV

PV effect is a fundamental physical mechanism that efficiently transforms light energy into electrical energy. These (I–V) characteristics of PV are given in Fig. 2 [19].

From Fig. 2, given as

$$I = I_L - I_D - I_{sh} \tag{1}$$

From Shockley diode equation, given as

$$I_D = I_o\left[\exp\left(\frac{V + IR_s}{nV_t}\right) - 1\right] \tag{2}$$

here,

$$V_t = \frac{kT}{q} \tag{3}$$

From the Ohm law, given as

$$I_{sh} = \frac{V + IR_s}{R_{sh}} \tag{4}$$

Substituting Eqs. (2)–(4) into Eq. (1), we have

$$I = I_L - I_o\left[\exp\left(\frac{V + IR_s}{nV_t} - 1\right)\right] - \frac{V + IR_s}{R_{sh}} \tag{5}$$

PV module total output current is given as

$$I = N_pI_L - N_pI_o\left[\exp\left(\frac{V/N_s + IR_s/N_p}{nV_t}\right) - 1\right] - \frac{V \times N_p/N_s + IR_s}{R_{sh}} \tag{6}$$

Where I = PV module output current (A), I_L = Light generated current (A), I_D = Diode current (A), I_{sh} = Leakage shunt current (A), I_o

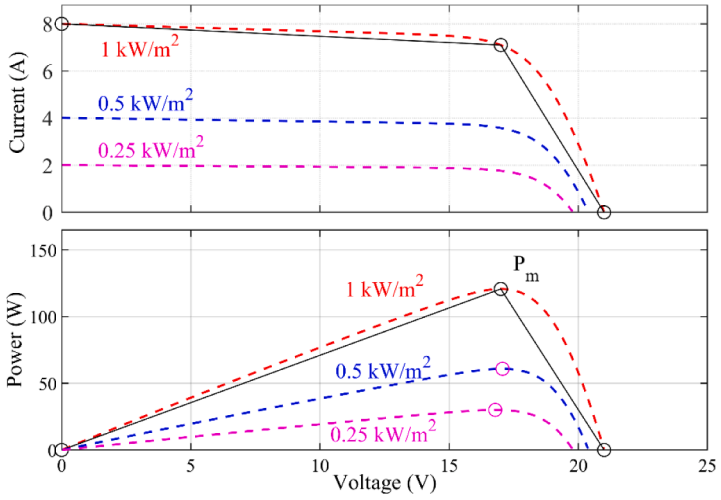
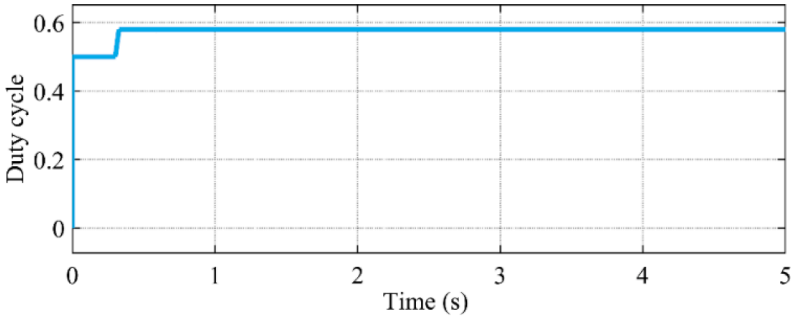
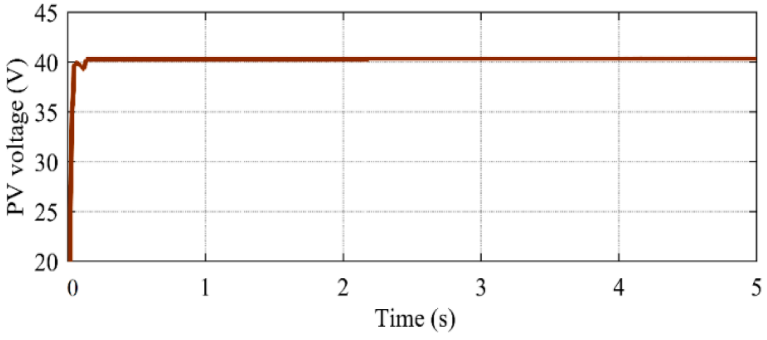


Fig. 3. I-V and P-V curve of one PV module.



(a)



(b)

Fig. 4. Simulation results for PV module at standard test condition ($1000 W/m^2 and 25^0 C$) (a) Duty cycle (b) PV voltage.

= Module saturation current (A), n = Ideality factor of diode, q = Electron charge ($1.6 \times 10^{-19} C$),R_{sh} = Shunt resistance (Ω), R_s = Series resistance (Ω),V = PV output voltage (V), k = Boltzmann's constant($1.3805 \times 10^{-23} J/K$), T = Absolute temperature (^oC), V_t = Thermal voltage ($0.0259 V at 25^oC$), N_p = Total number of PV cell connected in parallel, and N_s = Total number of PV cell connected in series.

Fig. 3 shows the P-V and I-V curve for one PV module. Fig. 4 show the simulation result for the PV module at the standard test condition($1000 W/m^2 and 25^0 C$). PV voltage is regulated by MPPT algorithm to realise this maximum power, and its value maintains approximately 20 kHz. In addition, PI controllers precisely regulate the output voltage and balance around its reference value.

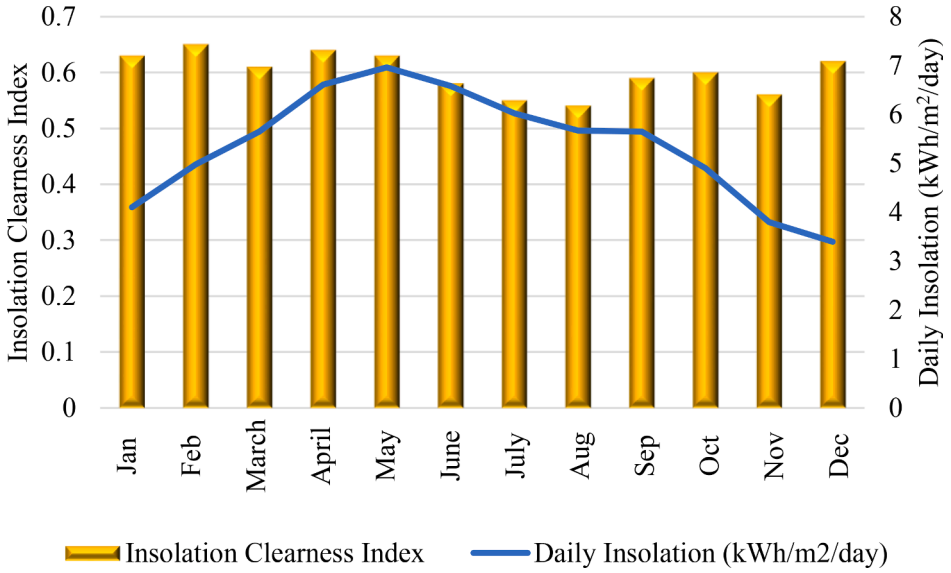


Fig. 5. Solar radiation and clearness index at (29.9476° N, 76.8155° E).

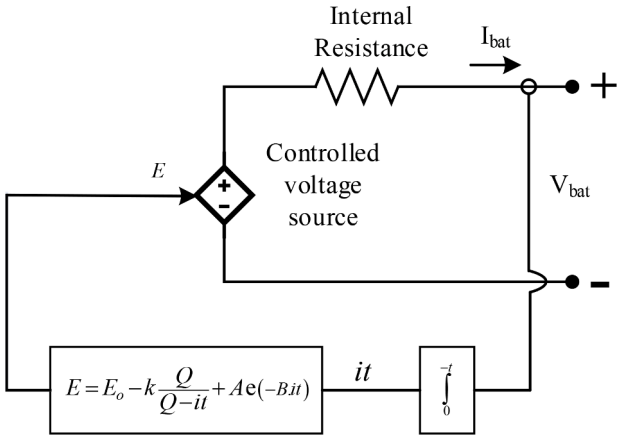


Fig. 6. Battery equivalent circuit diagram.

At a Kurukshetra location in India (29.9476° N, 76.8155° E), data on solar radiation have been acquired from NASA's Surface Meteorology and Solar Power database. The region had mean solar radiation of 5.36 kWh/m^2/day and an approximate clearness index of 0.60, as Fig. 5 shows.

2.2. *Mathematical modeling of battery*

The battery is connected in series with constant resistance and an internally operated voltage source in Fig. 6. Equations characterise the controlled voltage source [20]. The constant current/constant voltage (CC/CV) charging algorithm is well perfected and the most used in charging the lithium-ion battery. A constant current is employed to charge the battery in CC/CV charging method until the battery voltage rises to the maximum voltage. The charging voltage is kept constant, and accordingly, the charging current exponentially decreases. If given batteries ideal current is small enough, the charging process will stop, as Fig. 7 shows.

Charge equation ($i^* < 0$)for Li-Ion battery is expressed as

$$f_2(it, i*, i) = E_o - K \cdot \frac{Q}{it + 0.1 \cdot Q} \cdot i* - K \cdot \frac{Q}{Q - it} \cdot it + A \cdot \exp(-B \cdot it) \tag{7}$$

Discharge equation($i^* > 0$),

$$f_1(it, i*, i) = E_o - K \cdot \frac{Q}{Q - it} \cdot i* - K \cdot \frac{Q}{Q - it} \cdot it + A \cdot \exp(-B \cdot it) \tag{8}$$

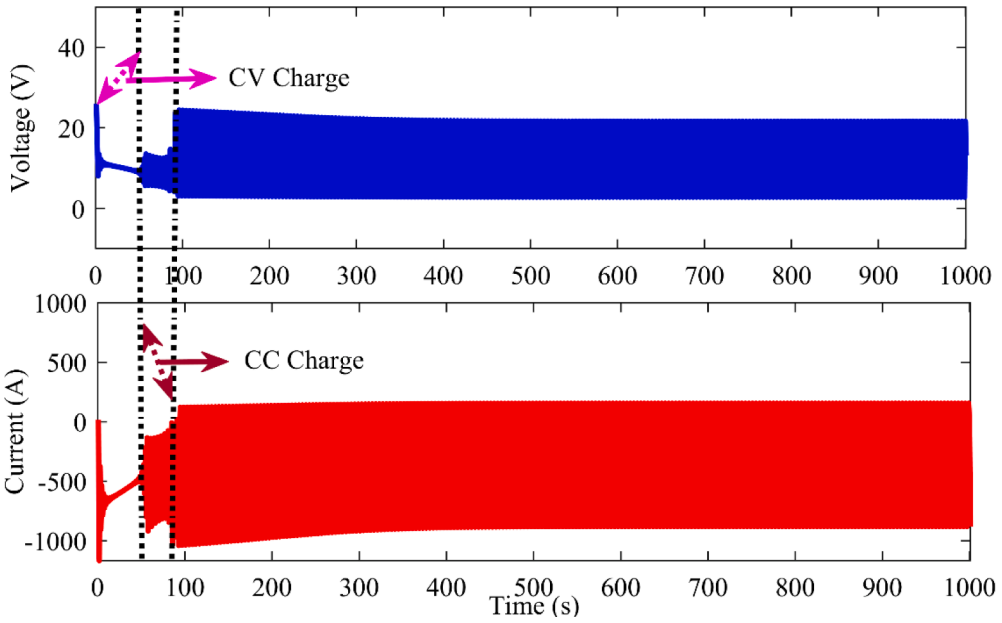


Fig. 7. CC/CV charging process.

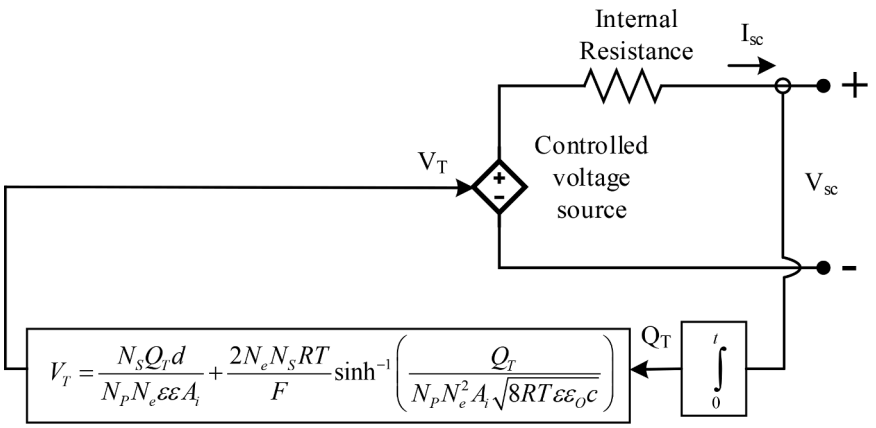


Fig. 8. Equivalent circuit of SC.

where E = Battery voltage at no load condition (V), E_o = Constant emf of battery (V), K = Polarization resistance of battery (Ω), i^* = Low-frequency current dynamics of battery (A), i = Terminal current of battery (A), Q = Maximum capacity of battery (Ah), k = Boltzmann's constant, A = Battery exponential voltage (V), it = Extracted capacity of battery (Ah),B = Battery exponential capacity (Ah^{-1}), andExp = Battery exponential zone dynamics (V).

2.3. Mathematical modeling of SC

Many models of SC characterise the electrical behaviour of SC [21]. The output voltage of SC (Fig. 8) uses the stern equation as

$$V_{SC} = \frac{N_S Q_T d}{N_P N_e \varepsilon\varepsilon_o A_i} + \frac{2N_e N_S RT}{F}\sinh^{-1}\left(\frac{Q_T}{N_P N_e^2 A_i\sqrt{8RT\varepsilon\varepsilon_O c}}\right) - R_{SC}\cdot i_{SC} \tag{9}$$

with

$$Q_T = \int i_{SC} dt \tag{10}$$

The supercapacitor's electric charge has been modified to be presented as self-discharge concept, when $i_{SC} = 0$,

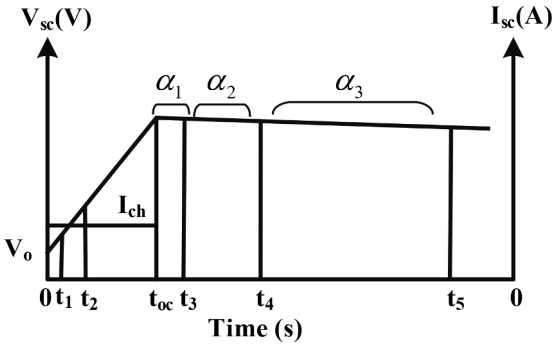


Fig. 9. Time versus SC voltage plot.

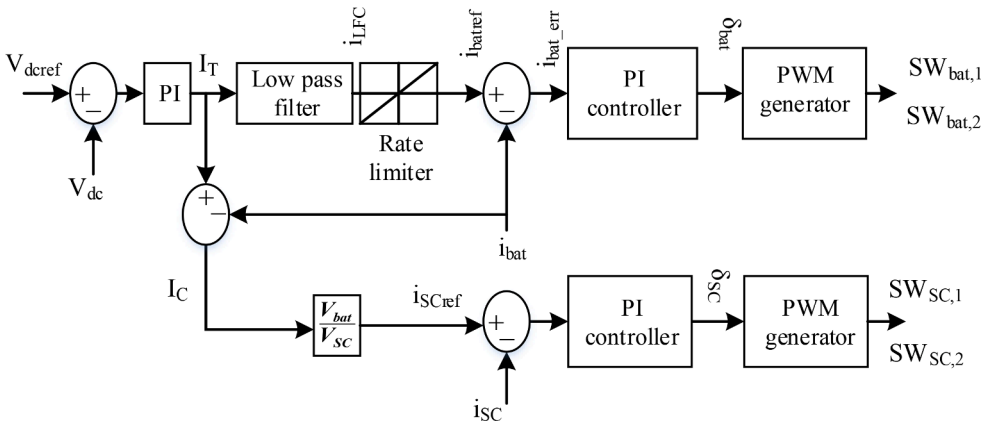


Fig. 10. Proposed hybrid energy storage system control strategy.

$$Q_T = i_{self_dis}dt \tag{11}$$

where

$$i_{self_dis} = \begin{cases} \dfrac{C_T\alpha_1}{1+sR_{SC}C_T} \; if \; t - t_{oc} \le t_3 \\ \dfrac{C_T\alpha_2}{1+sR_{SC}C_T} \; if \; t_3 \prec t - t_{oc} \le t_4 \\ \dfrac{C_T\alpha_3}{1+sR_{SC}C_T} \; if \; t - t_{oc} \succ t_4 \end{cases}$$

Where V_{sc} = SC output voltage (V), N_s = Number of series SC, N_p = Number of parallel SC, Q_T = Electric charge (Coulomb), N_e = Number of layer of the electrodes, d = Thickness of Helmholtz layer (meter), R = Ideal gas constant, F = Faraday constant, A_i = Interfacial area among electrodes and electrolyte ($meter^2$), T = Absolute temperature (^oK), c = Molar concentration ($mol/meter^3$), ε = Permittivity of material (Farad/meter), and ε_o = Permittivity of free space (Farad/meter).

α_1, $\alpha_2$and $\alpha_3$are the SC rates of change of voltage during time instant(t_{oc},t_3), (t_3,t_4), and(t_4,t_5), respectively, as Fig. 9 shows.

3. Control method for DCMGs

3.1. Control scheme for HESSs

Fig. 10 shows the proposed control method for DCMGs with HESSs. In Fig. 10,V_{dcref},i_{scref} and i_{batref} are the reference dc bus voltage, SC current, and battery current, respectively. i_{sc},i_{bat},I_T, $i_{LFC}$and $i_{bat_err}$are the SC, battery, total, low frequency component and battery error component current, respectively. The variables $\delta_{bat}$and δ_{sc} are duty ratios of battery and SC. The control method aims to monitor the DC bus voltage during differing power generation and load changes and generate reference current for converters.

The controller specifications have been tailored for the SC's high charge and discharge rates, so the controller responds accordingly [22]. SC internal current loop bandwidth is chosen f_{sw}/6to alleviate uncertainties in the study. f_{sw} is the switching frequency. The

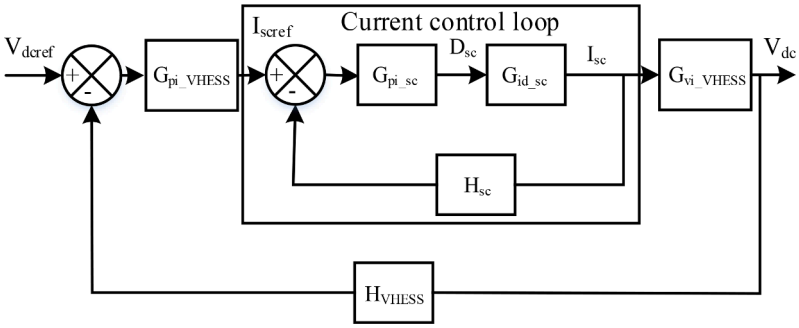


Fig. 11. Block diagram for SC converter.

Table 1
Nominal parameters of DCMGs.

Parameters	Specifications
PV array at standard test condition	$V_{OC} = 22.1\ V$, $I_{SC} = 8.7\ A$, $V_m = 17.6V$, $I_m = 6.8A$
Battery (Li-ion)	$12\ V$, $14Ah$ (2 Nos. connected in series)
Supercapacitor	$29\ F$, $32\ V$ (2 Nos. connected in series)
DC-DC converter	$L_{pv} = 11.4\ mH$, $L_{bat} = 1.2\ mH$, $L_{sc} = 1.8mH$, $C = 440\ \mu F$, $R = 20\ \Omega$

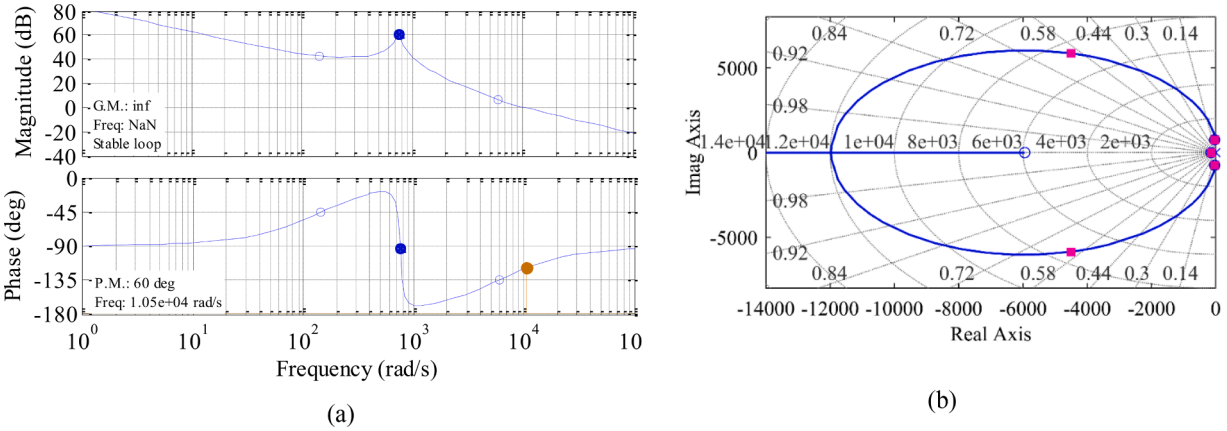


Fig. 12. (a) Bode plot of SC current controller (b) Locus of SC current controller.

current loop bandwidth of the battery is kept lower than SC i.e.$f_{sw}/20$ allowing for the instant increase of SC current. The bandwidth of the converter is accordingly lower than SCs' right half plane zero frequency (f_{rhpz}) [23] for safe operations:

$$f_{rhpz} = \frac{(1 - D_{SC})^2 R}{2\pi L_{SC}}, \tag{12}$$

where D_{sc} = SC duty ratio, R = Resistance(Ω), and L_{sc} = Supercapacitor filter inductance (H).

Voltage Control Loop (VCL) bandwidth is decided below the current control loop (CCL) SC because the VCL response is slower than the CCL.

3.1.1. Design of CCL

Fig. 11 shows the block diagram of CCL for the SC converter. The variables G_{pi_VHESS}, G_{vi_VHESS}, $G_{pi_sc}$and $G_{id_sc}$are the HESS voltage control loop compensator transfer function, HESS BDDC control-to-output transfer function, SC current control loop compensator transfer function, and supercapacitor BDDC control-to-inductor-current transfer function, respectively.H_{VHESS} is the HESS voltage sensor feedback gain and $H_{sc}$is the SC current sensor feedback gain. The small-signal changes in the duty ratio of SC converter($\widetilde{d}_{sc}$)to inductor current of SC($\widetilde{i}_{sc}$)transfer function (TF)$G_{cisc}(s)$ [24] is provided as

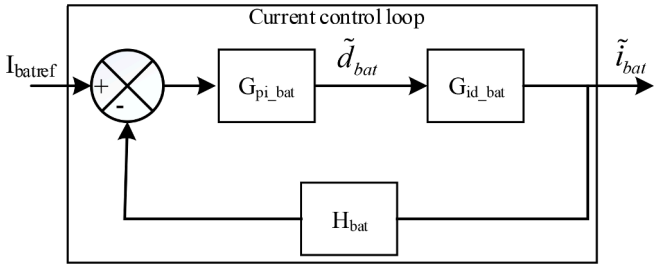
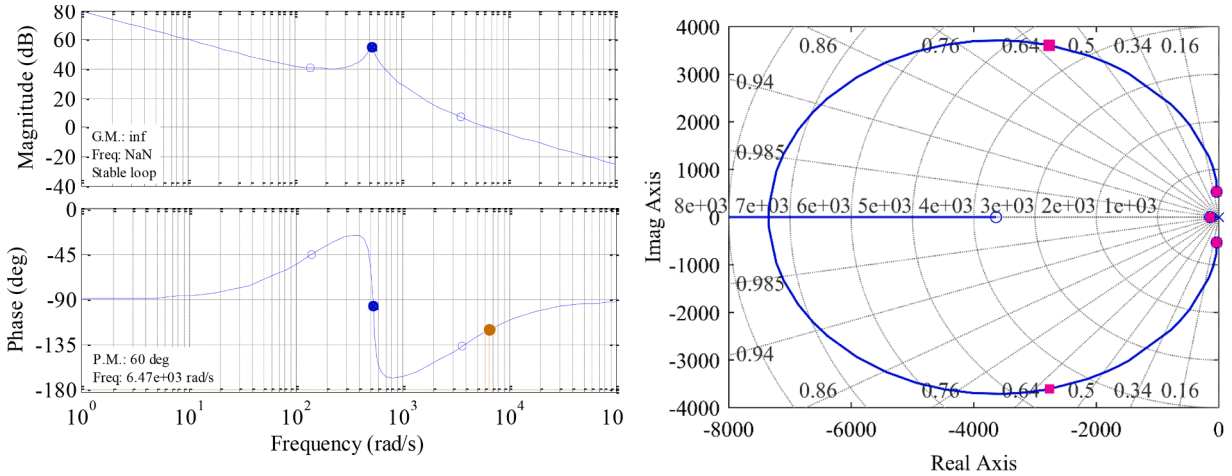


Fig. 13. Block diagram for battery CCL.



(a) (b)

Fig. 14. (a) Bode plot of battery current controller (b) Locus of battery current controller.

$$G_{id_sc} = \frac{\tilde{i}_{sc}}{\tilde{d}_{sc}} = \frac{V_{dc}Cs + 2\frac{V_{dc}}{R}}{L_{sc}Cs^2 + \frac{L_{sc}}{R}s + (1 - D_{sc})^2} \tag{13}$$

where G_{id_sc} = Supercapacitor TF of control-to-inductor-current supercapacitor, C = capacitance (μF), s = Laplacian, R = Resistance (Ω), L_{sc} = Inductance (mH), and D_{sc} = Duty ratio.

CCL compensator TF is provided as

$$G_{pi_sc} = K_{p_sc} + \frac{K_{i_sc}}{s}, \tag{14}$$

where G_{pi_sc} = TF of SC CCL compensator, K_{p_sc} = Gain of SC PI compensator and K_{i_sc} = Gain of SC PI compensator.

The open-loop TF of SC current controller is provided as

$$G_{openloop_sc} = G_{pi_sc}G_{id_sc}H_{sc}, \tag{15}$$

where H_{sc} symbolises the current sensor feedback gain.

Table 1 gives the SC converter parameters. Fig. 12(a,b) show the CCL bode plot and the root locus of the controller. PI compensator is used to compensate for the phase margin of 60^oat $1.05\,krad/$sand to determine the necessary values of $K_{p_sc} = 0.32$and $K_{i_sc} = 1974$. where K_{p_sc} = Gain of SC PI compensator and K_{i_sc} = Gain of SC PI compensator.

Fig. 13 shows the CCL block diagram for the battery. Likewise, small-signal changes in battery converter duty ratio($\tilde{d}_{bat}$)to inductor current of battery ($\tilde{i}_{bat}$) TF $G_{cibat}(s)$[24] is provided as

$$G_{id_bat} = \frac{\tilde{i}_{bat}}{\tilde{d}_{bat}} = \frac{V_{dc}Cs + 2\frac{V_{dc}}{R}}{L_{bat}Cs^2 + \frac{L_{bat}}{R}s + (1 - D_{bat})^2} \tag{16}$$

where G_{id_bat} = Battery TF of control-to-inductor-current, L_{bat} = Inductance (mH), and D_{bat} = Battery duty ratio.

The TF of CCL of the PI controller is shown as

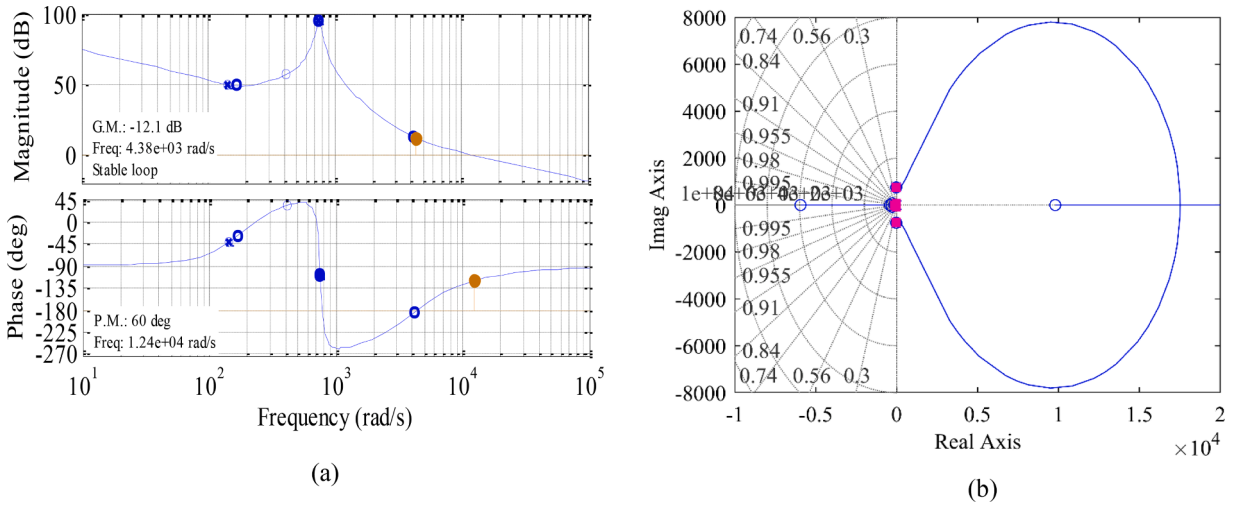


Fig. 15. (a) Bode plot for HESS voltage control loop (b) Root locus of voltage control loop of HESS.

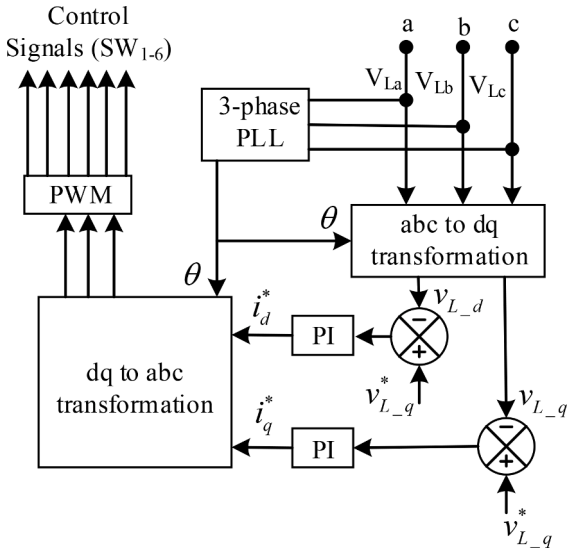


Fig. 16. Schematic structure of three-phase inverter controller.

$$G_{pi_bat} = K_{p_bat} + \frac{K_{i_bat}}{s}, \tag{17}$$

where G_{pi_bat} = Battery current control loop compensator TF, K_{p_bat} = PI compensator gain of battery, andK_{i_bat} = PI compensator gain of battery.

The battery current open-loop transfer function is described as

$$G_{openloop_bat} = G_{pi_bat} G_{id_bat} H_{bat}, \tag{18}$$

where H_{bat} symbolises the current sensor feedback gain.

Table 1 gives the battery converter parameters. Fig. 14(a,b) show the bode plot and root locus of CCL with compensator. PI compensator is used to compensate for the phase margin of 60^oat $6.47krad/s$and to determine the necessary values of $K_{p_bat} = 0.16$and $K_{i_bat} = 810$.

3.1.2. Design of VCL

Fig. 11 shows the SC converter block diagram for VCL. The small-signal changes in the SC inductor current $(\widetilde{i}_{sc})$ to DC bus voltage $(\widetilde{v}_{dc})$ TF $G_{visc}(s)$ [24] is provided as

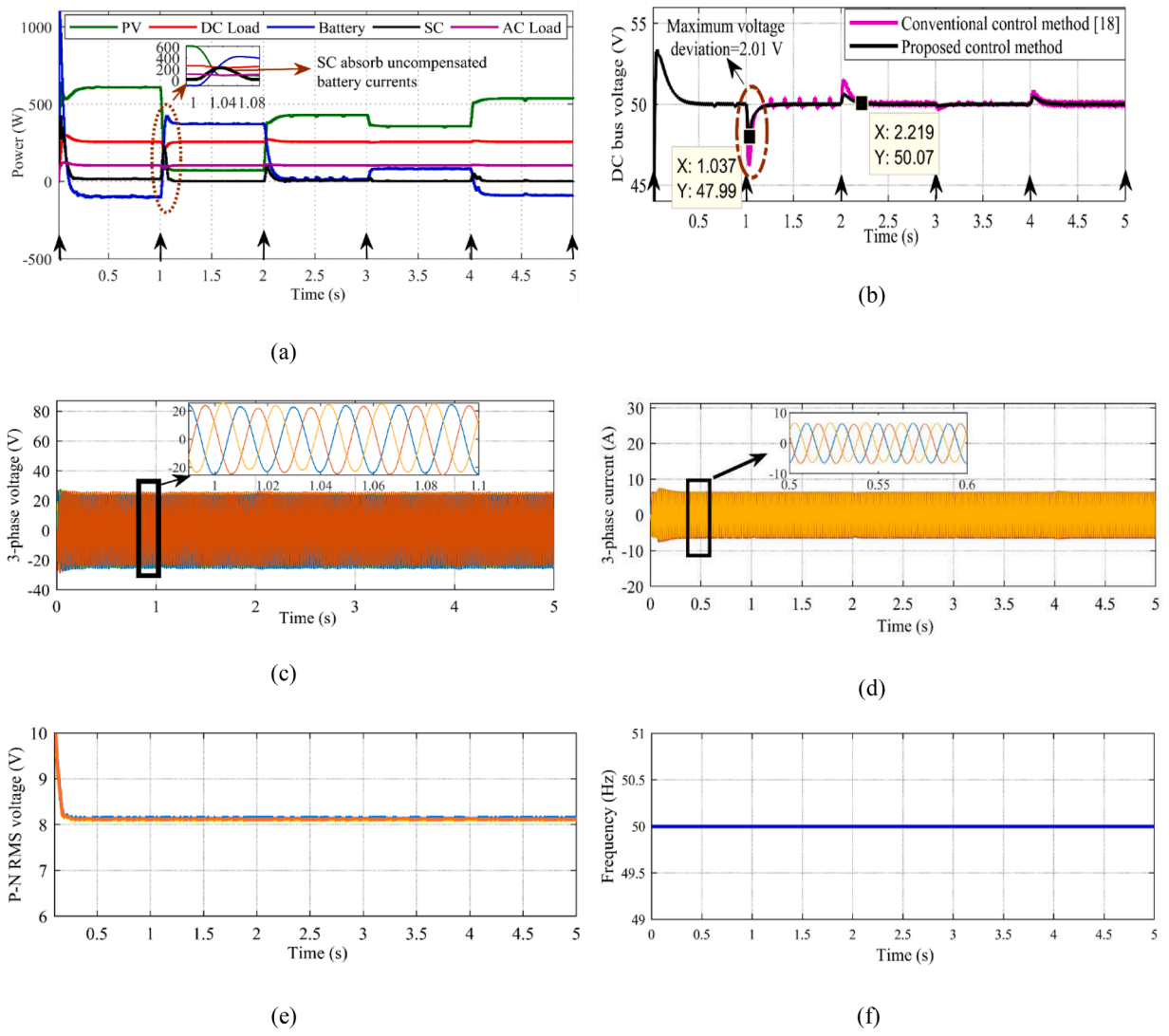


Fig. 17. Simulation results for variation in PV generation (a) Power-sharing (b) DC bus voltage (c) three-phase voltage via load (d) three-phase current via load (e) Phase-neutral root mean square voltage across load (f) Frequency.

The VCL TF is given as

$$G_{vi_VHESS} = \frac{\widetilde{v}_{dc}}{\widetilde{i}_{sc}} = \frac{R(1-D_{sc})\left(1-\frac{L_{sc}}{R(1-D_{sc})^2}s\right)}{2+RCs} \tag{19}$$

where G_{vi_VHESS} = BDDC control-to-output TF.

The VCL compensator TF is given as

$$G_{openloop_VHESS} = G_{pi_VHESS}G_{closeloop_sc}G_{vi_VHESS}H_{VHESS}, \tag{20}$$

where G_{pi_VHESS} = Voltage control loop TF, $G_{closeloop_sc}$ = Closed loop transfer function of SC, and G_{vi_VHESS} = Voltage control loop TF.

$$G_{closeloop_sc} = \frac{G_{pi_sc}G_{id_sc}}{1+G_{pi_sc}G_{id_sc}H_{sc}}, \tag{21}$$

where H_{VHESS} symbolises voltage sensor feedback gain.

Fig. 15(a,b) show the bode plot and root locus of VCL with compensator. PI compensator is used to compensate for the phase margin of 60^oat $1.24\,krad/$sand to determine the necessary values of K_{p_sc} = 1.68and K_{i_sc} = 652 .

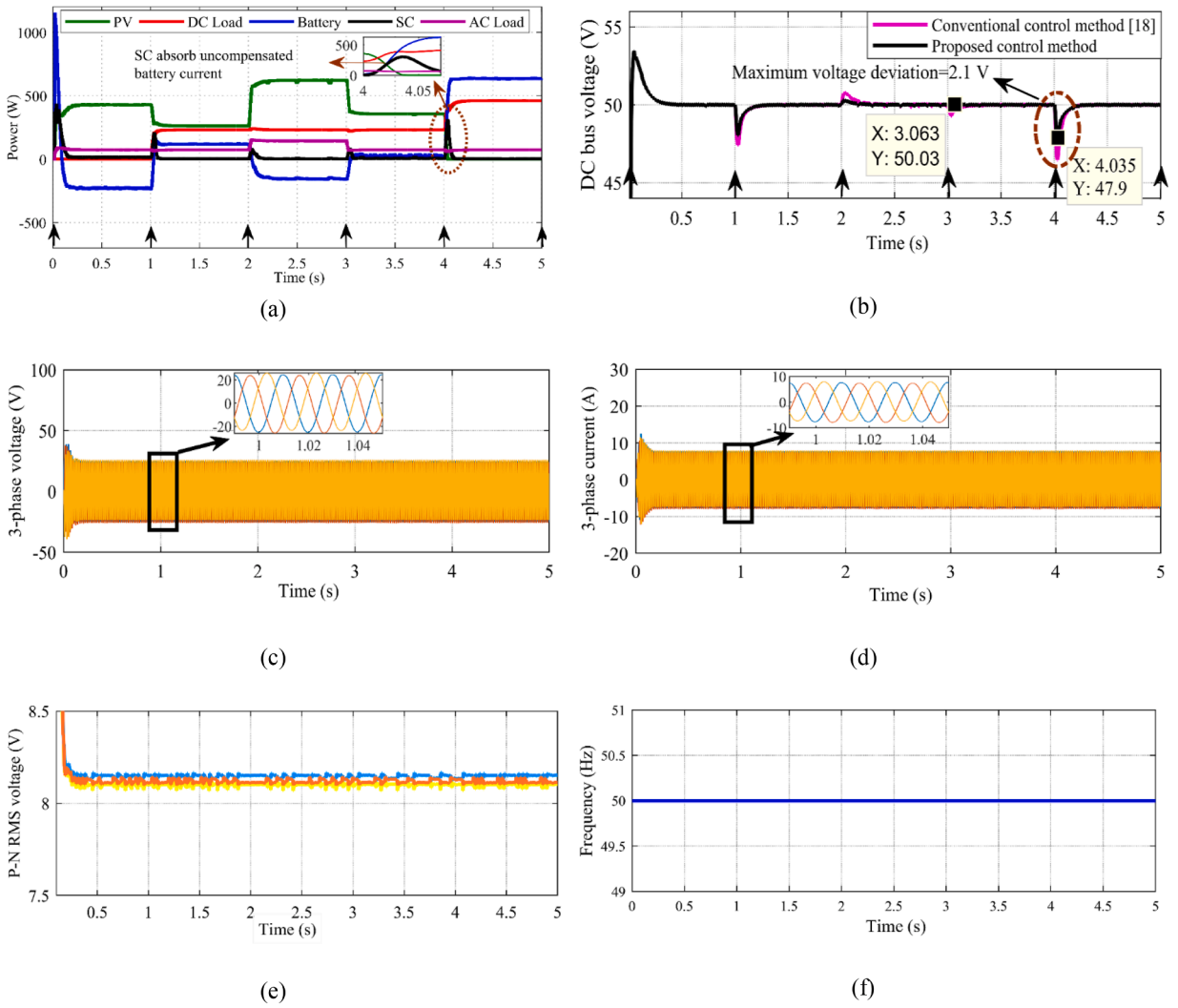


Fig. 18. Simulation results for change in load (a) Power-sharing (b) DC bus voltage (c) three-phase voltage via load (d) three-phase current via load (e) Phase-to-neutral root mean square voltage across load (f) Frequency.

3.2. Control strategy for three-phase inverter

The key function of a three-phase inverter is to transform defined DC bus voltage into a variable frequency and voltage. Fig. 16 presents the three-phase inverter controller producing the signals for switches. Vector control for the three-phase inverter controller, placed on the synchronously rotating reference point, is used for such a purpose. Eq. (22) expresses the dynamic voltage balancing equation through the filter of a three-phase inverter. Eq. (22) reveals the dynamic voltage equation [25] balancing equation through the filter of a three-phase inverter, as seen in Fig. 1.

$$\begin{bmatrix} v_{La} \\ v_{Lb} \\ v_{Lc} \end{bmatrix} = R_f \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + L_f \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix}, \tag{22}$$

where v_{La}, v_{Lb} and v_{Lc} are three-phase voltages via the load and i_a, i_b and i_c are the three-phase currents via the load, respectively. v_a, v_b and v_c symbolises the output voltages of three-phase inverter. R_f and L_f symbolises the filter resistance and inductance. These three-dimensional variables are transformed into synchronically rotating dimensions with angular velocity(ω) and are described as

$$v_{L_d} = v_d - R_f i_d - L_f \frac{di_d}{dt} + \omega L_f i_q, \tag{23}$$

$$v_{L_q} = v_q - R_f i_q - L_f \frac{di_d}{dt} - \omega L_f i_d, \tag{24}$$

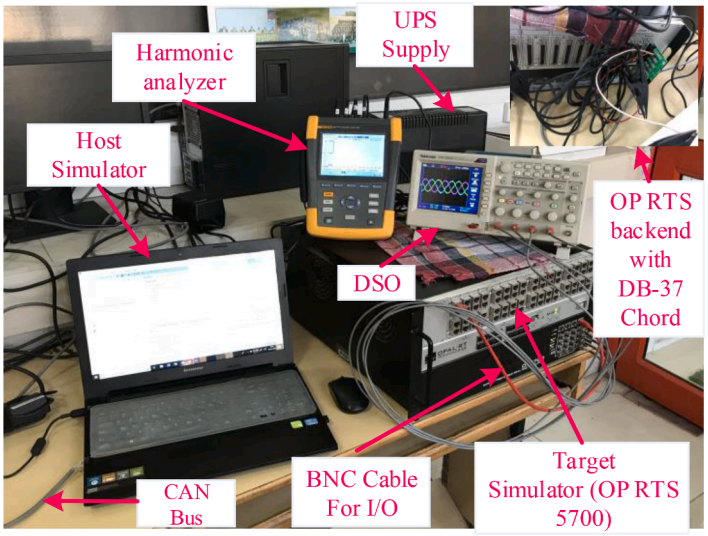


Fig. 19. Experimental set up.

where v_{L_d}, v_{L_q} symbolises the $(d-q)$component of AC voltages via load and i_d,i_q symbolises the $(d-q)$component of AC currents via load, respectively.

4. Result and discussion

4.1. Simulation results

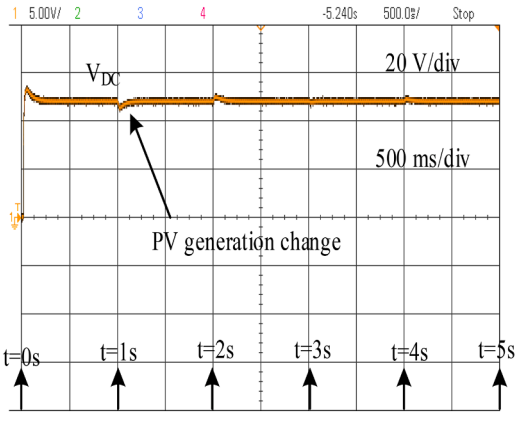
The DCMGs recognized seen in Fig. 1 are implemented in Matlab/SIMULINK. Table 1 depicts the rating of components of DCMGs. $V_{oc}$is the open circuit voltage(V), $I_{sc}$is the short circuit current(A), $V_m$is the voltage at maximum power (V)and $I_m$is the current at maximum power(A), respectively. Change in load and Variation in PV are the two schemes or events for which this proposed method is implemented.

4.1.1. Case-I variation in PV generation

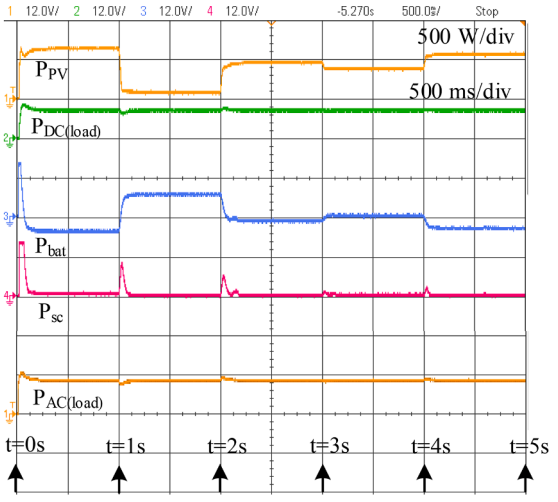
In this scenario, solar irradiance reduces DC bus voltage in PV power coherence. Here, the PV power has briskly dipped at $t = 1$–2 s and $t = 3$–4 s. The addition of DC Load and AC load is 392 W (load is constant), called the total load demand. At $t = 1$–2 s, PV power generation is 132 W, and during $t = 3$–4 s, the generation is 286 W. Battery energy storage systems and PV, together fulfills the load demand. The battery does not have dynamic stress as the SC is not imposing any on it because SC acquires a high-frequency part, and the battery is discharging. Output voltage is increased by PV power ratio at $t = 4$–5 s and $t = 0$–1 s as boost in solar irradiance increase PV power. 390 W is the total load demand, whereas 397 and 653 W are the total PV generation. PV systems meet the load demand, which infers that SC acquired high-frequency part and the battery is charging, and the battery receives the slow frequency element current. Due to this, the battery does not have dynamic stress, as depicted in Fig. 17(a). Fig. 17(b) depicts that the suggested control method permits swift and better DC bus voltage regulation. The battery's minimum dynamic stress is also due to the low charging rate of the battery. For example, if variations occur on the AC side in PV generation, the AC inverter controller delivers an adequate command to tackle this system's variations. The inverter circuit maintains a constant frequency, and hence reference frequency for a phase-locked loop (PLL) is supplied to it (Fig. 17(f)). Fig. 17(e) depicts phase-to-neutral (P-N) root mean square (RMS) voltage across the load, respectively. Fig. 17(c) shows the zoomed view of three-phase instantaneous voltage, and Fig. 17(d) shows current across the load, which admits minor variation when a change in PV generation occurs at $t = 1$ s.

4.1.2. Case-II change in load

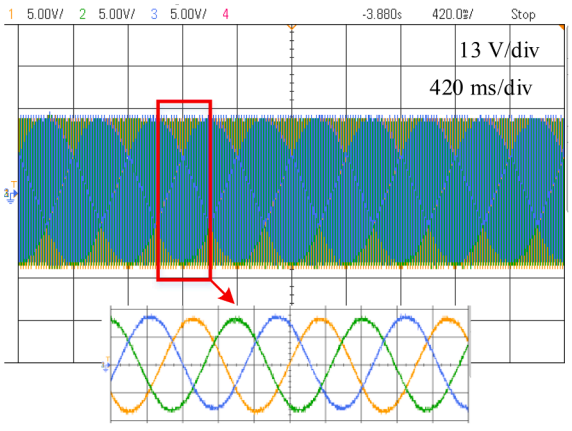
In this scenario, 12 Ω drop in resistance from 20 to 8 Ω increases the load demand quickly at time $t = 1$–2 s, and $t = 4$ s. Fig. 18(b) displays the decreased output dc bus voltage as load changes. Fig. 18(b) demonstrates that the suggested control method allows for quicker and improved DC bus voltage regulation. In Fig. 18(a), the battery does not have dynamic stress as the SC is not imposing any on it because SC acquires a high-frequency part and has managed the dc bus voltage at 50 V. At $t = 1$–2 s, PV power generation output power is 266 W, and during $t = 4$–5 s, the generation is 0 W is lower than 307 and 531 W load demand because the battery meets the load demand. At instant $t = 0$–1 s and $t = 2$–3 s, load demand of 73.6 and 371 W is actually less than the 433 and 620 W PV generation. Batteries can charge at this instance as PV meets load demand. Fig. 18(c) presents the waveform of three-phase voltage across load during load demand change. If variations arise on the AC side in the load, the AC inverter controller delivers an adequate command to tackle this system's variations. A virtual three-phase PLL has been used to produce theta angle to maintain the input and output signals



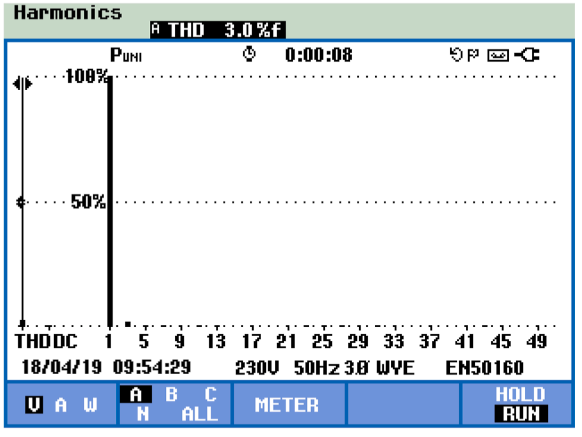
(a)



(b)



(c)



(d)

Fig. 20. Experimental results for variation in PV generation (a) DC bus voltage (b) Power-sharing (c) three-phase voltage (d) THD.

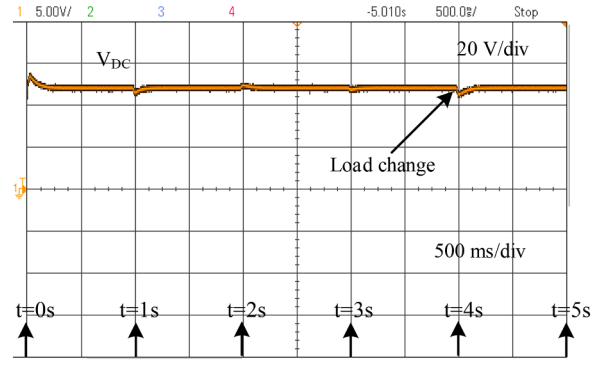
synchronized and maintain the grid-independent DCMG's constant frequency operation. Reference frequency for PLL is supplied to the inverter circuit, which maintains a constant frequency (Fig. 18(f)). Fig. 18(e) depicts P-N RMS voltage across the load, respectively. Fig. 18(c,d) reveal the zoomed view of three-phase instantaneous voltage and current via the load, which admit minor variation when a change in load occurs at $t = 1$, 3 and 4 s.

4.2. *Experimental results*

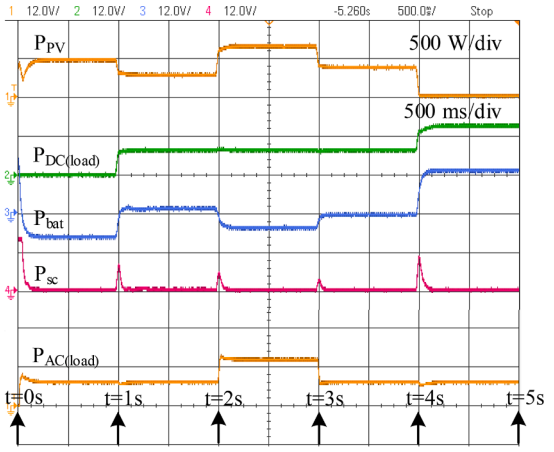
OPAL-RT is a field-programmable gate array-based real-time simulator on which the assessment of the suggested control method for DCMGs with HESSs is performed. Fig. 19 depicts the experimental setup of the proposed control method to authenticate its sturdiness. Setup includes a UPS supply, host and target simulator (OP 5700 RTS), harmonic analyzer, and mixed-signal Oscilloscope.

Fig. 20(a,b) depict experimental outcomes for DC bus voltage and power-sharing amongst HESSs for PV generation fluctuations. V_{dc} changes considerably for few PV generation disturbances, shown in Fig. 20(a). Fig. 20(b) explains the load gets power from the battery at time $t = 1$–2 s and $t = 3$–4 s because PV output power decreases due to solar irradiance reduction. However, the battery charges at time $t = 3$–4 s as the load gets power from PV generation output. For the variation in PV generation, the battery does not have dynamic stress as the SC is not imposing any on it because SC acquires a high-frequency part, which leads to an improved lifetime of the battery. The experimental result for 3-phase voltage is shown in Fig. 20(c). Following the IEEE std.519–1992 limit, the experimental result for 3-phase voltage confirms that THD for AC load is 3%, shown in Fig. 20(d).

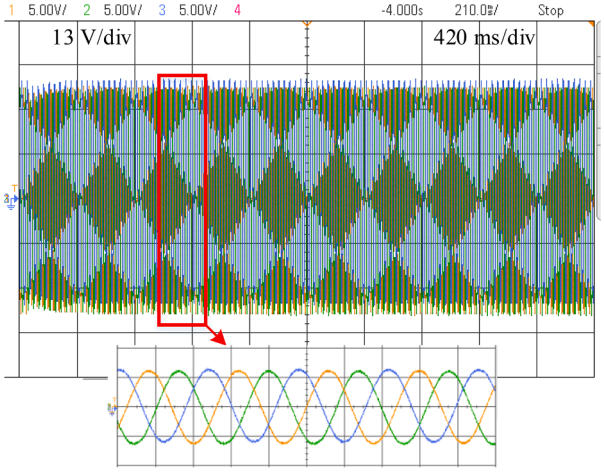
Fig. 21(a,b) present the experimental results for change in load. V_{dc} changes considerably for few disturbances in load, as shown in Fig. 21(a). Load demand is fulfilled by battery because load demand exceeds PV generation at time $t = 1$–2 s and 4–5 s, so the battery is



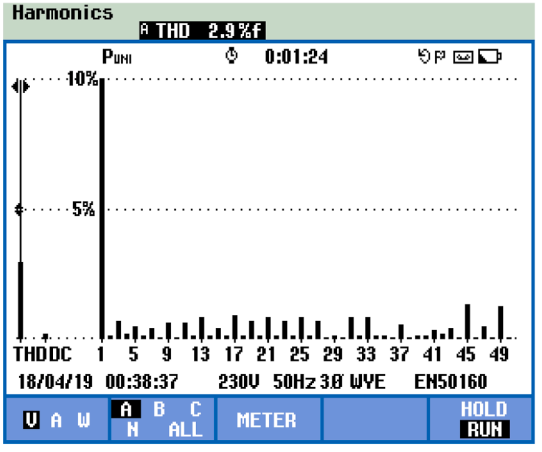
(a)



(b)



(c)



(d)

Fig. 21. Experimental results for change in load (a) DC bus voltage (b) Power-sharing (c) Three-phase voltage (d) THD.

Table 2
Performance comparison among conventional and proposed control methods.

Parameters			Conventional control method [18]	Proposed control method
DC bus voltage (V_{dc})	Peak overshoot/undershoot(M_P)	Case: I	3.95 V	2.09 V
		Case:II	4.01 V	2.1 V
	Settling time (t_{ss})	Case: I	390 ms	219 ms
		Case: II	179 ms	63 ms

discharging. At time $t = 0$–1 s and 2–3 s, load demand met by PV generation, so the battery gets to charge. The Experimental result for three-phase voltage is shown in Fig. 21(c). Fig. 21(d) confirms that THD for AC load is 2.9%, following the IEEE std.519–1992 limit.

4.3. *Performance comparison among conventional and proposed control method*

In the case-I and case-II mentioned above, the proposed control method's robustness has acknowledged the settling time and voltage overshoot/undershoot to restore the dc bus voltage. Table 2 describes the comparison among conventional and proposed control methods. The results reveal that the proposed control method achieves quicker and more accurate dc bus voltage regulation and needs lesser settling time than the conventional control method.

5. Conclusion

The Comparison of the proposed control method with the conventional control method for two cases for variation in PV generation and changes in load was presented in this paper. The extensive experimental study was carried out with the proposed control method for the faster regulation of DC bus voltage and effective power-sharing among HESSs. The reason behind this improvement was that the SC manages the high-frequency component until the battery current becomes steady state. Furthermore, the DCMG stability with controllers designed using the proposed control method provided better gain and phase margins. Additionally, it regulates the DC bus voltage within the permissible limit ± 5%per IEEE std. 519–1992. Moreover, THD analysis of three-phase AC load was also performed.

CRediT authorship contribution statement

Prashant Singh: Conceptualization, Data curation, Visualization, Formal analysis, Writing – original draft. **J.S. Lather:** Conceptualization, Data curation, Visualization, Formal analysis, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Prashant Singh is a research scholar at Department of Electrical Engineering, National Institute of Technology Kurukshetra, India. His-current research interests include DC Microgrids, Renewable Energy Systems, and Energy Management and Control Issues in DC microgrids and Hybrid Energy Storage Systems.

J.S.Lather is currently working as a professor in Department of Electrical engineering, National Institute of Technology Kurukshetra, India. His-research interests include Robust Control, Time Delay Systems, Renewable Energy and Microgrid Control, and Machine Learning Algorithms and Applications.